CHAPTER 2

Introduction to Probability

2.1 Sample space and events

Definition 2.1.1. A *random experiment* is a procedure whose outcome cannot be predetermined.

Although the outcome of the experiment cannot be predetermined, the set of all possible outcomes is usually known beforehand. This set is referred to as a *sample space* of the experiment.

Definition 2.1.2. A *sample space* is a set of all possible outcomes a random experiment can take. It is usually denoted by S.

Each repetition of the experiment procedure is called a *trial* and gives rise to one and only one of the possible outcomes.

Example 2.1.1. Some examples of random experiments and their corresponding sample spaces.

(a) Tossing a coin: $S = \{H, T\}$.

(b) Tossing two dice: $S = \{(i, j) \mid i, j = 1, 2, 3, 4, 5, 6\}$.

(c) Suppose a light bulb is allowed to burn until it burns out. Then the lifetime of the bulb is recorded as: $S\{t \mid t \geq 0\}$.

(d) A geologist takes rock samples in a mine in order to determine the quality of the iron-ore to be mined. The analytical lab reports the proportion of iron in an ore. Thus, $S = \{p \mid 0 \leq p \leq 1\}$.

Definition 2.1.3. Any subset of the sample space, S is referred to as an *event* of S.

Let A be an event of the sample space S. We then say that A *has occurred* if the outcome of the trial is any element/member of A. An *elementary event* is an event with exactly one element. The empty set event is referred to as an *impossible event* whereas S is called the *certain event*.

Definition 2.1.4. Let A and B be events of a sample space S. Then A and B are said to be *mutually exclusive* if $A \cap B = \emptyset$.

Example 2.1.2. A salesperson, after calling on a client records the outcome:

- sale made (S);
- sale potential good (P);
- no sale made (N).

List the sample space if (i) two clients are visited, and (ii) three clients are visited.

(i) $S = \{SS, SP, SN, PS, PP, PN, NS, NP, NN\}$.

(ii) $S = \{SSS, SSP, SSN, PSS, \ldots, NNN\}$

Example 2.1.3. A party of five (5) runners, two two females and three males run along a highway in a single line.

(a) What is the sample space S?

(b) Find the subsets that corresponds to the events:

(i) U : a female is in the lead;

(ii) V : a male is at the back;

(iii) W : females are in the second and fourth positions.

2.2 Axioms of Probability

To try to get some hindsight into the concept of probability, consider a random experiment with a sample space S repeated infinitely. We assume that the experiment is carried out under the same exact conditions. Define an event A of the sample space S and let r be number of times A occurs from n trials. Then the probability of A, $\mathbb{P}(A)$ is defined as

$$\mathbb{P}(A) = \lim_{n\to\infty} \frac{r}{n} \tag{2.1}$$

Thus, $\mathbb{P}(A)$ is a limiting frequency of A. As the number of trials increases, the relative frequency of A occurring tends to get closer and closer to the 'true' probability.

Now, consider an experiment whose sample space in S. For each event A of the sample space, we define the the probability of A, $\mathbb{P}(A)$ to be a real number with the following properties:

AI. $0 \leq \mathbb{P}(A) \leq 1$ for all $A \subset S$

AII. $\mathbb{P}(S) = 1$

AIII. For any mutually exclusive events A and B, i.e. $A \cap B = \emptyset$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

2.2.1 Some useful theorems

Theorem 2.2.1. Let $A \subset S$. Then $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$.

Proof. From properties of set operations we have that

$$A \cup \bar{A} = S.$$

Thus from AII,

$$\mathbb{P}\big(A \cup \bar{A}\big) = \mathbb{P}\big(S\big) = 1.$$

Since $A \cap \bar{A} = \emptyset$, then by AIII we have that

$$\mathbb{P}\big(A \cup \bar{A}\big) = \mathbb{P}\big(A\big) + \mathbb{P}\big(\bar{A}\big) = 1$$

Hence

$$\mathbb{P}\big(\bar{A}\big) = 1 - \mathbb{P}\big(A\big).$$

Theorem 2.2.2. Let $A, B \subset S$. Then $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap \bar{B})$.

Proof. Note that we can write A as the union of the following mutually exclusive sets:

$$A = \big(A \cap B\big) \cup \big(A \cap \bar{B}\big)$$

Clearly,

$$\big(A \cap B\big) \cap \big(A \cap \bar{B}\big) = A \cap \big(B \cap \bar{B}\big) = \emptyset.$$

Therefore by using Axiom III,

$$\mathbb{P}\big(A\big) = \mathbb{P}\big(A \cap B\big) + \mathbb{P}\big(A \cap \bar{B}\big)$$

Notice that from the above theorem, we have $\mathbb{P}\big(A \cap \bar{B}\big) = \mathbb{P}(A) - \mathbb{P}\big(A \cap B\big)$

Theorem 2.2.3. Let $A, B \subset S$. Then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof. We can write $A \cup B$ as follows

$$A \cup B = \big(A \cap \bar{B}\big) \cup B$$

and clearly

$$\big(A \cap \bar{B}\big) \cap B = \emptyset.$$

Thus, $\big(A \cap \bar{B}\big)$ and B are mutually exclusive and by Axiom III

$$\mathbb{P}\big(A \cup B\big) = \mathbb{P}\big(A \cap \bar{B}\big) + \mathbb{P}(B).$$

Note that $\mathbb{P}\big(A \cap \bar{B}\big)$ is given in Theorem 2.2.2 and hence we have

$$\mathbb{P}\big(A \cup B\big) = \mathbb{P}(A) + \mathbb{P}\big(B\big) - \mathbb{P}\big(A \cap B\big)$$

Theorem 2.2.4. Let $A \subset B$. Then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Proof. If $A \subset B$ then we can write B as a union of two mutually exclusive sets as follows

$$B = A \cup \big(B \cap \bar{A}\big)$$

and by Axiom III

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}\big(B \cap \bar{A}\big)$$

By Axiom I, $\mathbb{P}\big(B \cap \bar{A}\big) \geq 0$ and thus

$$\mathbb{P}(B) \geq \mathbb{P}(A)$$

Put the other way round

$$\mathbb{P}(A) \leq \mathbb{P}(B)$$

Theorem 2.2.5. Let $A_1, A_2, \ldots, A_n$ be pairwise mutually exclusive events, i.e. $A_i \cap A_j = \emptyset$ for $i \neq j$. Then

$$\mathbb{P}\big(A_1 \cup A_2 \cup \ldots \cup A_n\big) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \ldots + \mathbb{P}(A_n)$$

Or equivalently in a more compact form

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{i=1}^{n} \mathbb{P}(A_i)$$

Proof. Left as an exercise for the students.

Example 2.2.1. Let $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B) = 0.6$ and $\mathbb{P}(A \cap B) = 0.3$. Find $\mathbb{P}(\bar{B})$, $\mathbb{P}(A \cap \bar{B})$ and $\mathbb{P}(A \cup B)$.

Example 2.2.2. Is it possible for events in the same sample space to have the following probabilities: $\mathbb{P}(A) = 0.8$, $\mathbb{P}(\bar{B}) = 0.6$ and $\mathbb{P}(A \cap \bar{B}) = 0.7$?

Example 2.2.3. Let A, B and C be the events of the sample space S. Suppose $\mathbb{P}(A) = 0.7$, $\mathbb{P}(B) = 0.8$, $\mathbb{P}(C) = 0.1$, $\mathbb{P}(A \cup B) = 0.9$ and $\mathbb{P}\big[(A \cup B) \cap C\big] = 0.0$. Depict the events

A, B and C on a Venn diagram and find the probability of the following events:

(a) $A \cap B$

(b) $A \cap C$

(c) $A \cup B \cup C$

(d) $\bar{B} \cap C$

(e) $\overline{A \cap B}$.

2.3 Equally Likely Outcomes

There is a wide variety of problems for which it is natural to assume that the elementary events of the sample space are equal likely to occur. That is, if there are N elementary events contained in S, then the probability of each and every elementary event must be $1/N$.

Now let A be some event in this scenario. Then A must consist of the union of elementary events, each with probability $1/N$. Therefore if we could determine the number of elementary events in A, then

$$\mathbb{P}(A) = \frac{n(A)}{N} \tag{2.2}$$

where $n(A)$ is the number of elementary events in A and clearly $n(S) = N$.

Example 2.3.1. 100 people bought tickets in a charity raffle. 60 of them bought the tickets because they supported the charity. 75 bought tickets because they liked the prize. No one who neither supported the charity nor liked the prize bought a ticket.

(a) What is the probability that the prize-winning ticket was bought by someone who liked the prize?

(b) What is the probability that the prize was won by someone who did not support the charity?

(c) What is the probability that the prize was won by some who both supported the charity and liked the prize?

Example 2.3.2. A pack of playing cards contains 52 cards, 13 belonging to each of the four suites `spades, hearts, diamonds` and `clubs`. Within each suite the 13 cards are labelled: `Ace, 2, ..., 10, Jack, Queen, King`.

Let D be the event that a randomly selected card is a diamond, K be the event that the card selected is a King, and N be the event that the card selected has one of the numbers between 2 and 10. Find the following probabilities.

(a) $\mathbb{P}(D \cap K)$

(b) $\mathbb{P}(\bar{N} \cup D)$

(c) $\mathbb{P}(D \cup K \cup N)$

2.4 Combinatorial Probability

In real world applications, including cases where all elementary events are equally probable, it is usually impractical to list and/or count all the elementary events contained in the sample space. The theory of combinations and permutations enables us to compute the number of elementary events contained in a sample space and events quite easily.

2.4.1 Permutations

Consider the different ordered arrangements of the letters a, b and c. A complete enumeration shows that there are 6 different arrangements, namely `abc, acb, bac, bca, cab` and `cba`. Each arrangement is referred to as a *permutation*. Thus, there are 6 possible permutations of a set of 3 objects. But can't we generalize this for n objects?

A set containing n objects has

$$n(n-1)\cdot \ldots \cdot 3\cdot 2\cdot 1 = n! \qquad \text{read 'n factorial'}$$

different orderings of the objects in the set.

Example 2.4.1. A class in probability theory consists of 6 men and 4 women. An examination is given and the students are ranked according to their performance. Assume hat no two students obtain the same score.

(a) How many different rankings are possible?

(b) If the men are ranked among themselves and also the women, how many different rankings are possible?

Permutations of n objects taken r at a time

The number of ways of arranging $r (0 < r \leq n)$ objects from a total of n distinguishable objects is given by

$$(n)_r = \frac{n!}{(n-r)!}$$

When $r = n$, we adopt the convention that $0! = 1$.

Permutations with repetitions

The number of permutations of n objects taken r at a time, allowing for repetition is

$$n \cdot n \cdot \ldots \cdot n = n^r.$$

Example 2.4.2. Two new computer codes are being developed to prevent unauthorized access to classified information. The first consist of any six digits from `0 -- 9`; the second consist of any three digits from 0 – 9 followed by any two letters between `A -- Z` excluding `I` and `O`.

(a) Which code is better at preventing unauthorized access in a single attempt?

(b) If both codes are implemented, with the first code followed by the second, what is the probability of gaining access in a single attempt?

Example 2.4.3. A housewife is asked to rank five brands of washing powder (`A, B, C, D, E`) in order of preference. Suppose that she actually has no preference and her ordering is arbitrary. What is the probability that:

(a) brand `A` is ranked first;

(b) brand `C` is ranked first and brand `D` is ranked second.

2.4.2 Combinations

Suppose we have an interest in determining the number of different groups of r objects from a total of n objects. For example, how many ways can we form groups of three from five items, `A, B, C, D` and `E`. There are five ways to select the first item, 4 ways to select the second item and lastly 3 ways to select the third item. Thus, there are $5 \cdot 4 \cdot 3$ ways of selecting three items when the order is relevant. However, since the order is irrelevant in groupings and for every group of 3 there are $3 \cdot 2 \cdot 1$ different orderings, it follows that the total number of groups that can be formed is

$$\frac{5 \cdot 4 \cdot 3}{3 \cdot 2 \cdot 1} = 10.$$

In general, $n(n-1)\ldots(n-r+1)$ represents the number of grouping r items from a total of n items when the order of selection is relevant, and each group of r items will have $r!$ different orderings, then the number of different groups of r items that can be formed from a set of n items is

$$\frac{n(n-1)\dots(n-r+1)}{r!} = \frac{n!}{(n-r)!r!}$$

This is denoted by $\binom{n}{r}$ and represent the number of possible combinations n objects taken r at a time.

Example 2.4.4. In how many ways can a 9 man work team be formed from 15 men?

Example 2.4.5. From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed? What if 2 of the mean are feuding and refuse to serve on the committee?

Multinomial coefficients

Consider a set of n distinct items. Suppose this set is to be divided into r distinct groups of respective sizes $n_1, n_2, \dots, n_r$ such that $\sum_{i=1}^{r} n_i = n$. Then the number of ways of choosing r combinations of sizes $n_1, n_2, \dots, n_r$ from a set of n items is given by

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1!\, n_2! \, \dots \, n_r!}.$$

The numbers $\binom{n}{n_1, n_2, \dots, n_r}$ are known as *multinomial coefficients.*

Example 2.4.6. How many different ways can we write the word `PROBABILITY`

Example 2.4.7. A committee of 12 is to be split into three sub-committees, having three, four and five members respectively. In how many ways can these sub-committees be formed?

Example 2.4.8. A group of 20 people is to travel in three vehicles seating 4, 6 and 10 people respectively. What is the probability that three friends travel on the same car?